

CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Advanced Subsidiary Level and GCE Advanced Level

MARK SCHEME for the October/November 2013 series

9702 PHYSICS

9702/41

Paper 4 (A2 Structured Questions), maximum raw mark 100

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

Cambridge is publishing the mark schemes for the October/November 2013 series for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level components and some Ordinary Level components.



CAMBRIDGE
International Examinations

Page 2	Mark Scheme	Syllabus	Paper
	GCE A LEVEL – October/November 2013	9702	41

Section A

- 1 (a) work done in moving unit mass from infinity (to the point) M1 A1 [2]
- (b) (i) gravitational potential energy = GMm/x
energy = $(6.67 \times 10^{-11} \times 7.35 \times 10^{22} \times 4.5) / (1.74 \times 10^6)$ M1
energy = 1.27×10^7 J A0 [1]
- (ii) change in grav. potential energy = change in kinetic energy B1
 $\frac{1}{2} \times 4.5 \times v^2 = 1.27 \times 10^7$
 $v = 2.4 \times 10^3 \text{ m s}^{-1}$ A1 [2]
- (c) Earth would attract the rock / potential at Earth('s surface) not zero / <0
/ at Earth, potential due to Moon not zero
escape speed would be lower M1 A1 [2]
- 2 (a) (i) N : (total) number of molecules B1 [1]
- (ii) $\langle c^2 \rangle$: mean square speed/velocity B1 [1]
- (b) $pV = \frac{1}{3}Nm\langle c^2 \rangle = NkT$
(mean) kinetic energy = $\frac{1}{2} m\langle c^2 \rangle$ C1
algebra clear leading to $\frac{1}{2} m\langle c^2 \rangle = (3/2)kT$ A1 [2]
- (c) (i) *either* energy required = $(3/2) \times 1.38 \times 10^{-23} \times 1.0 \times 6.02 \times 10^{23}$ C1
= 12.5 J (12J if 2 s.f.) A1 [2]
or energy = $(3/2) \times 8.31 \times 1.0$ (C1)
= 12.5 J (A1)
- (ii) energy is needed to push back atmosphere/do work against atmosphere M1
so total energy required is greater A1 [2]
- 3 (a) (i) any two from 0.3(0) s, 0.9(0) s, 1.50 s (allow 2.1 s etc.) B1 [1]
- (ii) *either* $v = \omega x$ and $\omega = 2\pi/T$ C1
 $v = (2\pi/1.2) \times 1.5 \times 10^{-2}$ M1
= 0.079 m s^{-1} A0 [2]
or gradient drawn clearly at a correct position (C1)
working clear (M1)
to give $(0.08 \pm 0.01) \text{ m s}^{-1}$ (A0)

Page 3	Mark Scheme	Syllabus	Paper
	GCE A LEVEL – October/November 2013	9702	41

- (b) (i) sketch: curve from ($\pm 1.5, 0$) passing through (0, 25)
reasonable shape (*curved with both intersections between*
 $y = 12.0 \rightarrow 13.0$) M1
A1 [2]
- (ii) at max. amplitude potential energy is total energy B1
total energy = 4.0 mJ B1 [2]
- 4 (a) (i) force proportional to product of (two) charges and inversely
proportional to square of separation M1
reference to point charges A1 [2]
- (ii) $F = 2 \times (1.6 \times 10^{-19})^2 / \{4\pi \times 8.85 \times 10^{-12} \times (20 \times 10^{-6})^2\}$ C1
 $= 1.15 \times 10^{-18} \text{ N}$ A1 [2]
- (b) (i) force per unit charge M1
on *either* a stationary charge
or a positive charge A1 [2]
- (ii) 1. electric field is a vector quantity
electric fields are in opposite directions
charges repel
Any two of the above, 1 each B2 [2]
2. graph: line always between given lines M1
crosses x-axis between 11.0 μm and 12.3 μm A1
reasonable shape for curve A1 [3]
- 5 (a) (i) field shown as right to left B1 [1]
- (ii) lines are more spaced out at ends B1 [1]
- (b) Hall voltage depends on angle M1
either between field and plane of probe
or maximum when field normal to plane of probe
or zero when field parallel to plane of probe A1 [2]
- (c) (i) (induced) e.m.f. proportional to rate
of change of (magnetic) flux (linkage)
(*allow rate of cutting of flux*) M1
A1 [2]
- (ii) e.g. move coil towards/away from solenoid
rotate coil
vary current in solenoid
insert iron core into solenoid
(*any three sensible suggestions, 1 each*) B3 [3]

Page 4	Mark Scheme	Syllabus	Paper
	GCE A LEVEL – October/November 2013	9702	41

- 6 (a) force due to magnetic field is constant B1
force is (always) normal to direction of motion
this force provides the centripetal force A1 [3]
- (b) $mv^2 / r = Bqv$ M1
hence $q / m = v / Br$ A0 [1]
- (c) (i) $q / m = (2.0 \times 10^7) / (2.5 \times 10^{-3} \times 4.5 \times 10^{-2})$ C1
 $= 1.8 \times 10^{11} \text{ C kg}^{-1}$ A1 [2]
- (ii) sketch: curved path, constant radius, in direction towards bottom of page M1
tangent to curved path on entering and on leaving the field A1 [2]
- 7 (a) *either* if light passes through suitable film / cork dust etc. M1
diffraction occurs and similar pattern observed A1
or concentric circles are evidence of diffraction (M1)
diffraction is a wave property (A1) [2]
- (b) (speed increases so) momentum increases M1
 $\lambda = h/p$ so λ decreases M1
hence radii decrease A1 [3]
(special case: wavelength decreases so radii decreases – scores 1/3)
or
(speed increases so) energy increases (B1)
 $\lambda = h / \sqrt{2Em}$ so λ decreases (M1)
hence radii decrease (A1)
- (c) electron and proton have same (kinetic) energy C1
either $E = p^2 / 2m$ *or* $p = \sqrt{2Em}$ C1
ratio = $p_e / p_p = \sqrt{(m_e / m_p)}$ C1
 $= \sqrt{(9.1 \times 10^{-31}) / (1.67 \times 10^{-27})}$
 $= 2.3 \times 10^{-2}$ A1 [4]
- 8 (a) energy to separate nucleons (in a nucleus) M1
separate to infinity A1 [2]
- (b) (i) fission B1 [1]
- (ii) 1. U: near right-hand end of line B1 [1]
2. Mo: to right of peak, less than 1/3 distance from peak to U B1 [1]
3. La: 0.4 $\rightarrow$ 0.6 of distance from peak to U B1 [1]

Page 5	Mark Scheme	Syllabus	Paper
	GCE A LEVEL – October/November 2013	9702	41

- (iii) 1. right-hand side, mass = 235.922 u
mass change = 0.210 u C1 A1 [2]
2. energy = mc^2
 $= 0.210 \times 1.66 \times 10^{-27} \times (3.0 \times 10^8)^2$ C1
 $= 3.1374 \times 10^{-11} \text{ J}$ C1
 $= 196 \text{ MeV}$ (need 3 s.f.) A1 [3]
(use of 1 u = 934 MeV, allow 3/3; use of 1 u = 930 MeV or 932 MeV, allow 2/3)
(use of 1.67×10^{-27} not 1.66×10^{-27} scores max. 2/3)

Section B

- 9 (a) operates on / takes signal from sensing device B1
(so that) it gives an voltage output B1 [2]
- (b) thermistor and resistor in series between +4 V line and earth M1
 V_{OUT} shown clearly across *either* thermistor or resistor A1
 V_{OUT} shown clearly across thermistor A1 [3]
- (c) e.g. remote switching
switching large current by means of a small current
isolating circuit from high voltage
switching high voltage by means of a small voltage/current
(any two sensible suggestions, 1 each to max. 2) B2 [2]
- 10 (a) pulse (of ultrasound) B1
produced by quartz / piezo-electric crystal (1) B1
reflected from boundaries (between media) B1
reflected pulse detected B1
by the ultrasound transmitter (1) B1
signal processed and displayed (1)
intensity of reflected pulse gives information about the boundary (1)
time delay gives information about depth (1)
(four B marks plus any two from the four, max. 6) B2 [6]
- (b) shorter wavelength B1
smaller structures resolved / detected (*not more sharpness*) B1 [2]
- (c) (i) $I = I_0 e^{-\mu x}$ C1
ratio = $\exp(-23 \times 6.4 \times 10^{-2})$ C1
 $= 0.23$ A1 [3]
- (ii) later signal has passed through greater thickness of medium M1
so has greater attenuation / greater absorption / smaller intensity A1 [2]

Page 6	Mark Scheme	Syllabus	Paper
	GCE A LEVEL – October/November 2013	9702	41

- 11 (a) left-hand bit underlined B1 [1]
- (b) 1010, 1110, 1111, 1010, 1001
(5 correct scores 2, 4 correct scores 1) A2 [2]
- (c) significant changes in detail of V between samplings
so frequency too low M1
A1 [2]
- 12 (a) e.g. logarithm provides a smaller number
gain of amplifiers is series found by addition, (not multiplication)
(any sensible suggestion) B1 [1]
- (b) (i) optic fibre B1 [1]
- (ii) attenuation/dB = $10 \lg(P_2/P_1)$
 $= 10 \lg(\{6.5 \times 10^{-3}\}/\{1.5 \times 10^{-15}\})$
 $= 126$
length = $126 / 1.8$
 $= 70$ km C1
C1
A1 [3]